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Problem

PRACTICAL RAE GAPS

- Representation Autoencoders promise unified tokens for understanding and generation but reconstruct worse than specialized VAEs.
- Original RAE is incompatible with standard CFG and needs a second weaker model for AutoGuidance.
- Prior work treats only the encoder's final layer as the representation, ignoring information distributed across depth.

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Generalized RAE

SUM THE LAST K LAYERS

- Define the frozen encoder output as the sum of its last K layer features; K=1 recovers the original RAE.
- The latent footprint is unchanged and no encoder fine-tuning or specialized text/face data is required.
- Increasing K improves reconstruction monotonically; DINOv3-L at K=23 reaches rFID 0.18 and PSNR 27.03.

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RAE + REPA

- Across 27 encoders, RAE benefits mainly from global semantic quality, while REPA improves intermediate spatial structure.
- Using the same representation as encoder latent and intermediate alignment target helps rather than wastes capacity.
- This complementarity lets stronger encoders such as DINOv3-L produce the best generation results.

RAEv2

Improved Baselines with Representation Autoencoders

Faster Convergence, Better Reconstruction and Free Guidance | 2026

RAE performance is unlocked by three simple insights: aggregate multiple frozen encoder layers, combine RAE with complementary REPA alignment, and reuse the REPA head for zero extra post-training guidance.

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Self-guidance

REPA HEAD AS X-PREDICTOR

- In RAE latent space, the REPA projection head already predicts the clean representation x from an intermediate feature.
- Reparameterize the DiT output as x-prediction and use that early head as the weaker internal-guidance baseline.
- No second model and no unconditional CFG pass are needed, halving function evaluations relative to CFG.

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Evidence

10X FASTER CONVERGENCE

- RAEv2 reaches ImageNet-256 gFID 1.06 in 80 epochs and trains in about 10.5 hours on the reported setup.
- FDr6 improves from the prior 3.26 at 800 epochs to 2.17 at 80 epochs without post-training.
- EPFID@2 drops from 177 epochs for RAE to 35; gains also transfer to text-to-image and navigation world models.

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Meaning

- Frozen vision encoders preserve low-level detail across layers, not only high-level semantics at the top.
- Absolute gFID increments can hide impractical training cost, motivating epochs-to-quality as a standard metric.
- RAEv2 is a simplified baseline, not a new heavyweight architecture, making the recipe easy to adopt and extend.